Egg Segmentation & Tracking

CV Intern Assignment - Four-part solution

October 2026

What this report covers

Task 1	SAM ground-truth masks for 110 images using Grounding DINO + SAM
Task 2	Classical egg segmenter - 24 logged trials, leakage-safe split
Task 3	One-shot evaluation on the held-out split, full metric suite
Task 4	Two-egg video tracker with persistent IDs over 1 440 frames

Key results at a glance

Dev IoU (mean / median)	**0.612 / 0.758**	*87 images*
Eval IoU (mean / median)	**0.379 / 0.417**	*23 images, unseen scenes*
Dev instance F1 @ 0.5	**0.613**	
Eval instance F1 @ 0.5	**0.354**	
Tracker IDs over 1 440 frames	**2 (stable)**	*no false third ID*

Links

Tracked video (Drive): https://drive.google.com/file/d/1BbzyooMvdoqHriA8DdoZCqx427-M0TEk/view?usp=sharing

AI tool usage

I used an AI coding assistant (Google Antigravity / Gemini) to help write and debug Python code - fixing off-by-one errors in the evaluation script, debugging the tracker's Hungarian-matching logic, and reorganising the folder structure. All experiment numbers, trial decisions, and metric values come from real runs on the actual data. I did not generate or guess any numbers; the experiment log records every trial in the order it was run.

Task 1 - Ground-truth masks with SAM

Dataset

Roboflow egg v1 (110 images, CC BY 4.0). Images are 640×640 JPEG, folder-only classification labels, no bounding boxes or masks. The Roboflow split (77/22/11) was not used. Scenes include a single egg on ash, white-on-white tissue, a metal pot of water, a marble counter with three eggs, and a woven mat.

Pipeline

Grounding DINO (SwinT backbone, open-set text-prompted detection) was run with the prompt "egg" to produce bounding boxes. Each box was passed to SAM (ViT-H) as the prompt for a tight mask. The pipeline runs on Colab (A100). Output for each image: a combined binary mask and one mask per instance.

Manual review and fixes

Every one of the 110 overlays was inspected. Three images required hand corrections (documented in fixes.json): one dusty egg on ash where GDINO missed the box, one pot-rim segment that leaked into the mask, and one three-egg image where SAM merged two adjacent eggs. All fixes used a manual bounding box or foreground point re-prompt.

Output is committed at task1_sam/output/ so Tasks 2-4 can run CPU-only without re-running SAM.

Task 2 - Classical segmentation (24 trials)

Leakage-safe split

Images were embedded by border colour, perceptual hash (pHash), and filename-id proximity (nearby ids share a scene). Groups were clustered so whole scenes land on one side. The split is stratified to keep hand, ash, white-on-white, cloth, and multi-egg scenes on both sides.

Split	Images	Eggs (GT)
Dev	87	110
Eval	23	28

Algorithm overview (T24, final)

The algorithm runs at about 0.8 s per 640×640 image on a CPU. It was developed entirely on dev; eval was never touched during tuning.

- Resize to 320 px working size; bilateral denoise.
- Compute four candidate maps: L* channel (Lab), whiteness in HSV, background distance (difference from border colour in Lab), skin-suppressed map (YCrCb mask).
- Multi-scale threshold candidates at 10 levels; open with radii 1, 4, 8 px to break thin bridges to the background.
- Filter components by ellipse-like shape: area 2-65% of frame, solidity >= 0.82, aspect ratio <= 2.2, whiteness gate (mean white cue >= 0.5).
- Rank candidates by a composite quality score: shape^2 × (whiteness contrast + background contrast) × area^0.25 × (0.5 + edge support) × (1 - centre distance).
- Run GrabCut on each accepted egg using the candidate as the foreground seed.
- Upscale masks to original resolution; return all eggs above 40% of the top score.

Experiment log summary

24 trials were run in sequence. Each entry records the change, the reason, the parameters, the dev metrics, and the keep/discard decision. Failed trials stay in the log (EXPERIMENT_LOG.md). Key milestones:

Trial	Change	Result
T01	Otsu threshold baseline	IoU 0.306
T05	Lab + HSV candidate maps replace raw brightness	IoU 0.449
T08	Background-distance map added	IoU 0.497
T09	Skin suppression in YCrCb	IoU 0.523
T13	Ring-contrast + size ranking	IoU 0.568
T18	Multi-radius opening (unlocks hard cases)	oracle 0.68->0.76
T19	Edge + centre + shape ranking terms	IoU 0.593, median 0.758
T24	Whiteness gate on candidates (FINAL)	IoU 0.612, median 0.758

Discarded trials include: GrabCut ablation (T14), tighter GrabCut margin (T15), guarded GrabCut (T16), mean-shift candidate maps (T17), watershed egg splitting (T20), ellipse regularisation (T21), and two quality-threshold sweeps (T22, T23).

Task 3 - Evaluation on the held-out split

Protocol

The frozen T24 algorithm was run once on the 23 eval images. Ground truth is the Task 1 SAM masks. Pixel accuracy is not the headline metric (background dominates, giving 0.88/0.84 on dev/eval). The reported metrics are IoU, Dice, precision, recall, boundary F1 at 3-pixel tolerance, 95th-percentile Hausdorff distance, instance F1 matched at IoU 0.5, and absolute egg-count error.

Results

Metric	Dev mean	Dev median	Dev std	Eval mean	Eval median	Eval std
IoU	0.612	0.758	0.359	0.379	0.417	0.307
Dice	0.684	0.862	0.345	0.478	0.588	0.327
Precision	0.718	0.988	0.389	0.543	0.560	0.420
Recall	0.761	0.920	0.336	0.605	0.742	0.391
BF1 (3 px)	0.545	0.575	0.335	0.417	0.466	0.262
HD95 (px)	131	66	166	250	217	230
Instance F1 @ 0.5	0.613	1.000	0.475	0.354	0.000	0.416
Abs. count error	0.161	0.000	0.452	0.391	0.000	0.488

Dev: 87 images, 110 GT eggs, 100 predicted. 63% of images IoU >= 0.5; 53% IoU >= 0.7.
Eval: 23 images, 28 GT eggs, 25 predicted. 26% of images IoU >= 0.5; 17% IoU >= 0.7.

Scene-level breakdown

Scene	Images	Mean IoU
keyboard (dev)	4	0.990
leaves (dev)	5	0.922
paisley cloth (dev)	4	0.921
woven mat (dev)	3	0.868
hand dark wall (dev)	3	0.858
marble counter (dev)	15	0.822
steel pot (eval)	12	0.542
earphones carpet (eval)	3	0.561
blue bucket (eval)	2	0.185
white box (eval)	2	0.076
hand backlit (eval)	3	0.000
red stool (eval)	1	0.000

Failure analysis

Backlit egg - dark silhouette

Whiteness gate (white cue >= 0.5) was set because orange lids and blue bag print failed on dev. Back-lit eggs are dark silhouettes and are rejected by the same gate. All three hand_backlit images score IoU 0.00. Dev had only front-lit hands (mean IoU 0.86), so this corner never appeared during tuning.

Large pale surface outranks egg

The ranking multiplies shape by whiteness contrast and by area^0.25. A pale tile wall or a white card covers more area and scores higher than the egg. egg_031 and egg_032 output an empty mask because the winning blob (65% of the frame) is then rejected by the area limit. egg_030 keeps the card: recall 1.00, precision 0.15, IoU 0.15.

** GrabCut leaks into pale floor or lid*

When the egg is on a similar-coloured surface, GrabCut floods outward. recall stays near 1 while precision collapses (egg_149 IoU 0.25, egg_150 IoU 0.12, egg_156 IoU 0.16). A guarded GrabCut (T16) was tested on dev but discarded because the dev set did not contain this exact case.

** Dimmer egg in water not a candidate*

In the steel-pot images with two eggs, the brighter egg is segmented cleanly (precision 1.00) and the darker one never passes the whiteness gate (recall 0.35-0.48). Watershed splitting (T20) cannot recover an egg that was never a candidate.

** Boiling foam - ground truth is approximate*

egg_135 and egg_145 are submerged under boiling water. The SAM mask is a blob around the foam, noted at review time. Part of the boundary error is SAM's outline, not the algorithm's. egg_124 (clearer shell) scores 0.88.

Where it already works

Best eval: egg_159 (dry egg in pot, IoU 0.99), egg_153 (IoU 0.94), egg_124 (IoU 0.88), egg_062 (carpet, IoU 0.83). Best dev scenes: keyboard 0.99, leaves 0.92, paisley cloth 0.92, woven mat 0.87, marble counter 0.82 (multi-egg). The pattern is consistent: high-contrast egg on a darker or busier background -> reliable segmentation.

What would close the gap (not applied - eval untouched)

- Replace the absolute whiteness gate with a relative contrast gate: this fixes backlit eggs.
- Reject candidates covering > 40% of the frame before ranking: removes the tile wall and the white card.
- Limit GrabCut to pixels that differ from the image border by a minimum Lab distance: stops floor and lid leaks.
- After accepting one egg, search locally for a second ellipse: recovers the dim egg in water.

Task 4 - Video tracking

Video

"Raw and Boiled Egg Spinning" by MocomiKids (YouTube). Two eggs are spun on a clean table surface. The video is 1 440 frames at 24 fps. 640 frames show both eggs; 142 frames show one egg (the other is under a hand); 658 frames are title cards.

Why the still-image segmenter was not used

The T24 algorithm takes ~0.8 s per frame and its whiteness gate was tuned for photographs with complex backgrounds. The video background is a clean table; the challenges are motion blur, hand occlusion, a logo, and on-screen captions.

Video-specific detector

- Bright, low-saturation pixels: HSV saturation <= 60, value >= 165.
- Skin removal in YCrCb (removes the hand).
- Fixed-region blanking: strip above the table and the 'mocomi' logo corner.
- Component filter: area 1 400-14 000 px, solidity >= 0.82, aspect <= 1.9.
- Touching-egg split: distance-transform watershed when two peaks > 0.6 × max distance.

Tracker

Hungarian assignment on IoU + centroid distance between frames. A Kalman filter propagates each egg's position during occlusion. An ID is kept for up to 15 frames of occlusion before being retired. When the egg reappears it is re-matched by centroid proximity and the same ID is reattached.

Results

Metric	Value
Total unique IDs over 1 440 frames	2 (no false third ID)
Frames with both IDs visible	640
Frames with one ID (other occluded)	142
Frames with no ID (title cards)	658
ID stability through spin (fastest blur)	ID re-attached on first sharp frame
Reflections / logo / captions detected	None

Egg 1 starts on the left. A hand picks it up, puts it on the right, and it remains Egg 1 throughout. Egg 2 stays on the left. During the fastest spin, the moving egg is a motion blur and is not segmented for ~1 second; the ID is held by the Kalman filter and reattached as soon as the egg is sharp again.

Tracked video: https://drive.google.com/file/d/1BbzyooMvdoqHriA8DdoZCqx427-M0TEk/view?usp=sharing